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REMARKS/3	

1)Draw the ideal characteristics of all types of filters.

Sol:

The magnitude characteristics of lowpass, highpass and bandwidth are shown,

IDEAL LOW PASS FILTER

The ideal lowpass filter characteristics is defined by the transfer function H(jΩ) = $e^{-j\Omega t_d}$ $\Omega \leq |\Omega_c|$

= 0 $\Omega > |\Omega_c|$

Where Ω_c is called as cutoff frequency of the filter. We can find the impulse response of the lowpass filter by finding the inverse Fourier transform of eq. given below that is

H(jΩ) = | H(jΩ)| $e^{-j\Omega t_d}$ for an ideal filter, in passband |H(jΩ)|=1 therefore

H(jΩ) = $e^{-j\Omega t_d}$

h(t)=$F^{-1}[H(j\Omega)]$

$$= \frac{1}{2\pi}\int_{-\infty}^{\infty} \mathrm{H(j\Omega)}\, e^{-j\Omega t_d}\, d\Omega \quad = \frac{1}{2\pi}\int_{-\Omega_c}^{\Omega_c} e^{-j\Omega t_d}\; e^{j\Omega t}\, d\Omega$$

$$= \frac{1}{2\pi}\int_{-\Omega_c}^{\Omega_c} e^{j\Omega(t-t_d)}\, d\Omega \;= \frac{1}{2\pi j(t-t_d)}\; [\, e^{j\Omega(t-t_d)}\,]_{-\Omega c}^{\Omega c}$$

$$= \frac{1}{2\pi j(t-t_d)}[\, e^{j\Omega(t-t_d)} - \; e^{-j\Omega(t-t_d)}] = \frac{1}{\pi(t-t_d)}\left[\frac{e^{j\Omega(t-t_d)} - e^{-j\Omega(t-t_d)}}{2j}\right]$$

$$= \frac{1}{\pi(t-t_d)}\sin\big(\Omega c(t-t_d)\big) = \frac{\Omega_c}{\pi}\,\frac{\sin\big(\Omega c(t-t_d)\big)}{\big(\Omega c(t-t_d)\big)}$$

$$= \frac{\Omega_c}{\pi}\, sinc[\Omega c(t-t_d)]$$

The impulse response h(t) is sketched as shown in in Fig. 11.5. From Fig. 11.5. filter becomes an all pass filter, as t_{d} -> 0 the output response peak of is to be cutoff frequency. The width of the main lobe is $(2\pi)/ft_c$ Notice that as $\Omega c \to \infty$ response approaches input. Further we can observe that the impulse response is non zero for t < 0 even though the input delta(t) is applied at t = 0 That is the imp response begins even before the input is applied. In real life, no system exhibits type of characteristics. Hence we can conclude that the ideal low pass filter in fig is physically not realizable. Similarly we can show that other ideal filters like highpass and bandpass filters are also physically not realizable.

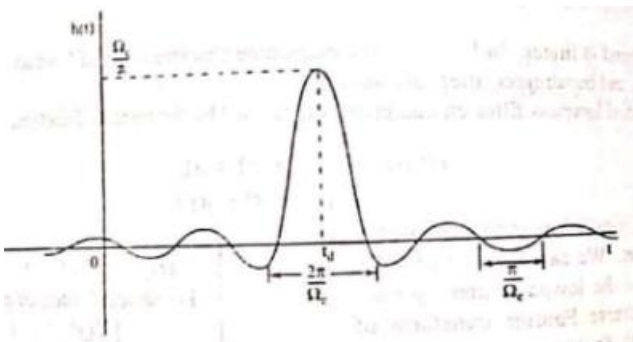


Fig. 11.5 The impulse response of an ideal lowpass filter.

From the above discussion, note that for a physically realizable system, h(t) = 0 causal, that is h(t) = 0 for t < 0

In the frequency domain, the condition for physical realizability is given by

$$\int_{-\infty}^{\infty} \frac{|\ln H(j\Omega)|}{1+\Omega^2}\, d\Omega < \infty$$

IDEAL HIGH PASS

A high-pass filter allows high-frequency signals to pass through and blocks low-frequency signals.

Ideal Characteristics:

1. Perfect passband:

All frequencies above the cutoff frequency (f_h) pass with constant gain (no attenuation).

→ Gain = 1 for f > f_h

2. Perfect stopband:

All frequencies below the cutoff frequency are completely attenuated (no output).

→ Gain = 0 for f <f_h

3. Sharp transition:

The change from stopband to passband occurs abruptly at the cutoff frequency (vertical slope).

4. No phase distortion:

The phase shift is linear with frequency in the passband (ideal case).

5. Infinite selectivity:

It perfectly separates low and high frequencies.

IDEAL BAND PASS FILTER

A band-pass filter allows a certain range of frequencies (between two cutoff frequencies) to pass and blocks others.

Ideal Characteristics:

1. Perfect passband:

Frequencies between lower cutoff (f_i) and upper cutoff

(f_h) pass without attenuation.

→ Gain = 1 for f_1 <f<f_h

2. Perfect stopbands:

Frequencies below fi and above f_h are completely attenuated.

→ Gain = 0 for f < for f > f_h

3. Sharp transition at both edges:

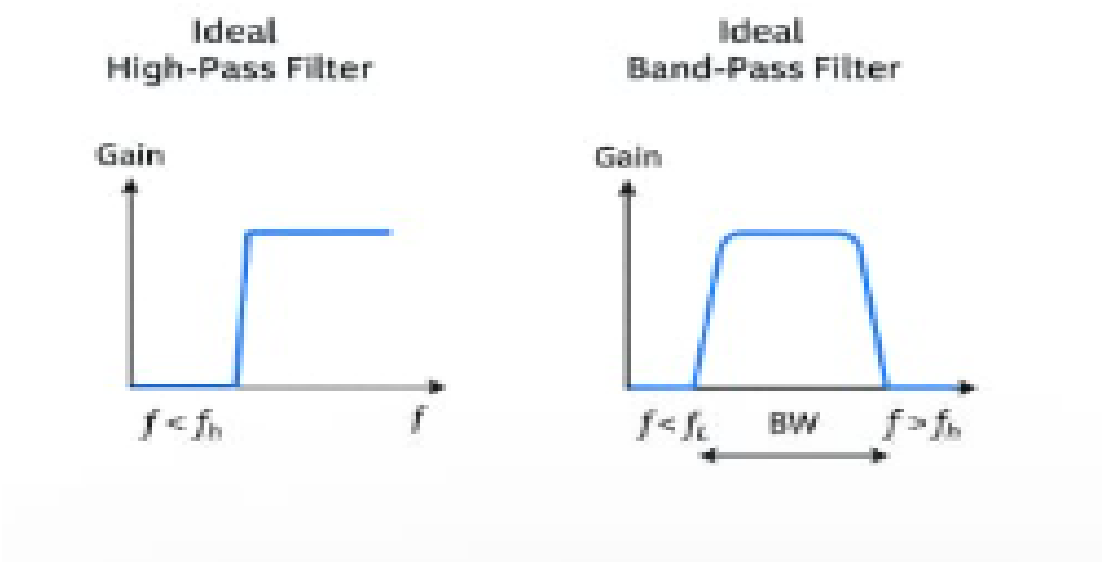
Instant change from 0 to 1 at the cutoff frequencies (ideal sharpness).

4. No distortion:

Linear phase response within the passband.

5. Bandwidth (BW):

BW = f_h-f_l ideally, only that range passes cleanly.



2) A continuous random variable X is uniformly distributed over the internal [0,2]. Calculate PDF , f(x), mean E(x).

Solution)

The rule of PDF is $\int_{-\infty}^{\infty} f_x(x) = 1$

$$E(x)= \int_{-\infty}^{\infty} x\, f_x(x)$$

Since PDF is uniformly distributed $f_x(x)$=constant

$$\int_{-\infty}^{\infty} f_x(x)dx = 1$$

$$\int_0^2 f_x(x)dx = 1$$

$$\int_0^2 c\, dx = 1$$

$$c\int_0^2 dx = 1$$

$$c[x]_0^2 = 1$$

$$c[2]=1$$

$$c=1/2$$

E(x)= $\int_0^2 x\frac{1}{2}dx = 1$

$=\frac{1}{2}\int_0^2 x\, dx =\frac{1}{2}\ [\frac{x^2}{2}]_0^2$

$= \frac{1}{2}[\frac{4}{2}]$**=1**

3) Derive the cross power spectral density of the signals X(t) and Y(t), and write the properties with proper mathematical illustrations.

Sol) Cross power Density Spectrum:

For a two real random processes X(t) and Y(t), we define as ensemble members; $X_t(t) and\ Y_t(t)$

that is

$$x_T(t) = \begin{cases} x(t) & -T < t < T \\ 0 & else \end{cases}$$

$$y_T(t) = \begin{cases} y(t) & -T < t < T \\ 0 & else \end{cases}$$

Both $x_t(t) and\ y_t(t)$ **are assumed to have bounded Vanation and to be magnitude integrable over the interval (-T, T),**

As a consequences, they will **posses Fourier transforms that are denoted by**

$X_T(\omega) and\ Y_T(\omega)$**" respectively.**

$$x(t) \leftrightarrow X_T(\omega)$$

$$y_T(t) \leftrightarrow Y_T(\omega)$$

Cross power $P_{XY}(T)$ **in the two process within the interval (-T,T) by**

$$P_{XY}(T) = \frac{1}{2\pi}\int_{-T}^{T} x_t(t)\, y_t(t)\ dt$$

$$=\frac{1}{2\pi}\int_{-T}^{T} x(t)\, y(t)\ dt$$

Since $x_t(t) and\ y_t(t)$ are Fourier transformable, parsevals theorem applies: then

$$P_{XY}(T) = \frac{1}{2T}\int_{-T}^{T} x(t)y(t)dt$$

$$= \frac{1}{2T}\int_{-\infty}^{\infty} \frac{X_T*(\omega)Y_T(\omega)}{2T} d\omega$$

Apply $\lim_{T\to\infty}$., Expectation to Calculate the total random process power, and replace x(t) by X(t).

$$P_{XY}(T) = \lim_{T\to\infty} \frac{1}{2\pi}\int_{-T}^{T} E[x(t)y(t)]dt$$

$$=\frac{1}{2\pi}\int_{-\infty}^{\infty} \quad \lim_{T\to\infty} \frac{E[X_T*(\omega)Y_T(\omega)]}{2T} dw$$

It is clear that the integral involving ω can be defined as a "Cross power density spectrum", it is a function of ω which we denote

$$S_{XY}(\omega) = \lim_{T\to\infty}.\frac{E[X_T*(\omega)Y_T(\omega)]}{2T}$$

Then we obtain the cross power formula

$$P_{XY} = \frac{1}{2\pi}\int_{-\infty}^{\infty} S_{XY}(\omega)\ d\omega$$

Cross power is given by The real part of By repeating the above procedure , we can also define another cross power density spectrum by

$S_{XY}(\omega) \lim_{T\to\infty}.\frac{E[X_T*(\omega)Y_T(\omega)]}{2T}$

$$P_{XY} = \frac{1}{2\pi}\int_{-\infty}^{\infty} S_{XY}(\omega)\ d\omega = P_{XY}\ *$$

Properties of Cross power density spectrum:

1) The real part of $S_{XY}(\omega)$ and the real part of $S_{XX}(\omega)$ are even function of ω
   Proof is $S_{XY} = \int_{-\infty}^{\infty} R_{XY}\ (\tau)e^{-j\omega t}dt$

   We know

$$e^{-jt\omega} = cos\omega\tau - jsin\omega\tau$$

$$R[S_{XY}(\omega)]= \int_{-\infty}^{\infty} R_{XY}\ (\tau)cos\omega\tau dt$$

$=\int_{-\infty}^{\infty} R_{XY}\ (\tau)\cos\ (-\omega\tau)d\omega$

Cos is even function $R[S_{XY}(\omega)]= R[S_{XY}(-\omega)]$

2) $S_{XY}(\omega) = 0$ **and** $S_{YX}(\omega) = 0$**. If X(t) and Y(t) are orthogonal.**
Proof is if X(t) and Y(t) are orthogonal then $R_{XY}(\tau) = R_{YX}(\tau) = 0$

$$S_{XY} = S_{YX} = 0.$$

3) $S_{XY}(\omega)$ and $R_{XY}(\tau)$ are f.T pairs.

Already proved in relation between cross correlation and cross power density spectrum.